

DeReFusion Ablation Study Report: Component Contributions and a Weight-Initialization Confound

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Abstract—This report documents a controlled component ablation of the DeReFusion forecasting framework. Three ablation variants—**woDy** (base branch only), **woLSTM** (Transformer-only residual), and **woTransformer** (LSTM-only residual)—are evaluated on two assets (S&P 500 index and BTC/USD) at horizon $T=24$, seed 2021, under the protocol of the accompanying reproduction study. Seven runs are reported: the three variants on each of two assets, plus a **revin-DLinear** environment-calibration run whose result (MSE 0.07007) matches the reference value exactly. Three findings stand out. First, the LSTM is the dominant component of the residual branch: removing it *improves* the model on GSPC (0.06150 vs. 0.06230) and the Transformer-only variant is the best of the three on both assets. Second, the **woDy** ablation is confounded: it differs from the **revin-DLinear** baseline not by removing a component but by dropping the DLinear weight-initialization convention, and this alone accounts for an 8% MSE gap at identical parameter count. Third, component value is asset-dependent: the full hybrid wins on BTC/USD while the LSTM-free variant wins on GSPC. All seven results are reproduced from the official code on CPU with a single seed.

Index Terms—financial time series forecasting, ablation study, DeReFusion, decomposition-residual architecture, weight initialization, reversible instance normalization

I. INTRODUCTION

DeReFusion combines a linear decomposition base branch (DLinear) with a learned LSTM-Transformer residual branch under direct additive fusion, wrapped in Reversible Instance Normalization (RevIN). The accompanying reproduction study establishes the pipeline and its central qualitative claim; this report isolates the question that follows immediately: which components of the residual branch actually carry the model, and what does each removal cost?

The study is deliberately narrow. It covers two assets (GSPC, BTCUSD), one horizon ($T=24$), one seed (2021), and three ablation variants, for seven executed runs in total. Within that scope the results are unambiguous and, in one case, counter-intuitive: the most complex ablation is not the best one, and one of the ablation variants is not a clean ablation at all.

The remainder is organized as follows. Section II defines the variants and the controlled protocol. Section III records the environment and the calibration run. Section IV presents the measured results. Section V interprets them, including the initialization confound. Section VI concludes and lists next steps.

II. ABLATION DESIGN

A. Variants

The repository ships three ablation models under `models/derefusion/ablation_variant/`. All three retain RevIN, the DLinear base branch, and direct additive fusion; they differ only in the residual branch or in its removal:

- 1) **woDy — base only**. The residual branch is removed entirely; the model reduces to RevIN-wrapped DLinear (4,664 parameters).
- 2) **woLSTM — Transformer residual**. The residual branch keeps a single Transformer encoder layer (4 heads, FFN $4d$, dropout 0.3) and drops the LSTM (19,988 parameters).
- 3) **woTransformer — LSTM residual**. The residual branch keeps the LSTM and drops the Transformer (11,988 parameters).

For reference, the full DeReFusion residual branch (LSTM $\rightarrow$ Transformer) has $\sim 28k$ parameters.

B. Controlled Protocol

Every run uses an identical protocol, unchanged from the reproduction baseline: chronological 70/10/20 split; Standard-Scaler fitted on the training split only; RevIN on every trained model; multivariate input with single-target output (MS; target Close); look-back $L=96$; label length 48; Adam, learning rate 10^{-4} , batch size 32, MSE loss; cosine schedule; early stopping with patience 5 on validation MSE; ~ 30 epochs; seed 2021; CPU execution. Metrics are MSE, MAE, RMSE, MAPE, MSPE, and R^2 , computed on the normalized series.

The grid is 3 variants $\times$ 2 assets = 6 runs, plus one calibration run (Section III), for seven runs total.

III. ENVIRONMENT AND CALIBRATION

All runs execute on a single Windows machine with a CPU-only PyTorch build, no WSL, no GPU. Table I lists the shared configuration.

a) Calibration run.: Following the receipt protocol, a **revin-DLinear** baseline with a known reference answer was executed first: GSPC, $T=24$, seed 2021. The measured MSE is **0.070065**, against a reference of 0.07007 — a difference of 6×10^{-6} , far inside the accepted tolerance band of 0.066–0.074. The environment is therefore confirmed correct, and

TABLE I
SHARED CONFIGURATION OF ALL ABLATION RUNS.

Parameter	Value
Look-back L / label len.	96 / 48
Horizon T	24
Input channels / target	4 (OHLC) / Close
d_{model}	32
Moving-average kernel k	25
Residual Transformer / LSTM	1 layer each (4 heads, FFN $4d$)
Dropout	0.3
Optimizer / learning rate	Adam / 1×10^{-4}
Batch size / loss	32 / MSE
Schedule	cosine
Early stopping	patience 5 (val. MSE)
Random seed	2021
Device	CPU

the ablation results below can be compared directly with the reference table.

IV. RESULTS

A. Measured Metrics

Table II reports the seven executed runs. All values are computed on the normalized series, consistent with the paper’s convention; the best MSE within each asset group is bold.

B. Comparison With the Full Model

Table III places the best ablation per asset against the full DeReFusion model and the linear baseline. The reference values for the full model and the baseline are reported by the collaborating study on the same data and protocol; they are reproduced here for comparison and are not re-executed in this work.

V. ANALYSIS AND FINDINGS

A. Finding 1: The LSTM Is Not the Component That Carries GSPC

The clearest result in Table II is that removing the LSTM improves GSPC: `woLSTM` reaches MSE 0.06150, better than the full model’s 0.06230 and better than every other variant. A three-point ordering emerges on GSPC:

$$\text{woLSTM} < \text{woTransformer} < \text{revin-DLinear} \approx \text{woDy} \quad (1)$$

Interpreted component-wise, the Transformer contributes positively to the residual branch (its removal costs 0.0075 MSE relative to `woLSTM`), while the LSTM contributes negatively — adding it back raises MSE slightly. On a near-random-walk equity index, the LSTM’s local sequential memory appears to fit noise rather than signal at this sample size.

This is a negative result and is reported as such: the ablation does not show that every component helps. On BTCUSD

the picture reverses (Section V-C), so the negative LSTM contribution is asset-specific rather than universal.

B. Finding 2: The `woDy` Ablation Is Confounded by Initialization

`woDy` is intended as the pure base-branch ablation, and by construction its forward computation is identical to the `revin-DLinear` baseline: RevIN normalize, moving-average decomposition, two linear maps, add, RevIN denormalize. Both models have exactly 4,664 parameters. Yet on GSPC they differ by 8% in MSE (0.07586 vs. 0.07007).

Inspection of the two implementations locates the cause. The `revin-DLinear` model follows the DLinear paper’s convention and initializes both linear layers to the uniform averaging kernel:

$$W_{\text{se}}, W_{\text{tr}} \leftarrow \frac{1}{L} \mathbf{1}_{T \times L}, \quad (2)$$

which makes the model start as a naive moving-average extrapolator. `woDy` uses PyTorch’s default Kaiming-uniform initialization for the same layers. Same architecture, same parameter count, different starting point — and a materially different optimum.

The practical consequence is that `woDy` is *not* a clean measure of the residual branch’s contribution: its 0.07586 gap to the full model mixes the removal of the residual branch with the loss of the initialization convention. A clean base-only ablation must reuse the DLinear initialization; we flag this as a correctness issue rather than report it as a component finding.

C. Finding 3: Component Value Is Asset-Dependent

On BTCUSD the ordering is different. `woLSTM` still leads the ablations, but now the full model wins: DeReFusion at 0.21797 beats `woLSTM` at 0.22178, which in turn beats the baseline at 0.22322. The LSTM, which hurt on GSPC, helps on BTCUSD — consistent with an asset whose dynamics are less random-walk-like and therefore carry learnable sequential structure.

Across both assets the Transformer-only variant is more stable than the LSTM-only variant, and the LSTM-only residual is consistently the weakest of the two residual designs where both are present. The residual branch’s value is thus real but asset-conditioned, which is exactly the regime that motivates structure-aware routing.

D. Parameter Efficiency

Parameter count does not track accuracy. `woLSTM` (19,988 parameters) beats `woTransformer` (11,988) and `woDy` (4,664) on both assets, but the ranking is not monotone in size: `woDy` matches the baseline’s parameter count while performing worse, because of the confound above. Within the residual designs, the larger variant wins on both assets.

TABLE II
ABLATION RESULTS, ALL RUNS EXECUTED LOCALLY ($T=24$, SEED 2021, CPU). BEST MSE PER ASSET GROUP IN BOLD. MAPE/MSPE IN PERCENT.

Asset	Model	MSE	MAE	RMSE	MAPE (%)	MSPE (%)	R^2	Params
GSPC	woLSTM	0.06150	0.18714	0.24799	5.80	0.651	0.8708	19,988
GSPC	woTransformer	0.06897	0.20742	0.26262	6.36	0.704	0.8551	11,988
GSPC	revin-DLinear	0.07007	0.21356	0.26470	6.52	0.712	0.8528	4,664
GSPC	woDy	0.07586	0.21988	0.27543	6.71	0.765	0.8406	4,664
BTCUSD	woLSTM	0.22178	0.36189	0.47094	9.32	1.504	0.8734	19,988
BTCUSD	woTransformer	0.22942	0.36869	0.47898	9.45	1.539	0.8690	11,988
BTCUSD	woDy	0.23727	0.37389	0.48711	9.58	1.583	0.8645	4,664

TABLE III
BEST ABLATION VS. REFERENCE FULL MODEL AND LINEAR BASELINE
($T=24$, SEED 2021). LOWER MSE IS BETTER.

Asset	Model	MSE	R^2
GSPC	woLSTM (local)	0.06150	0.8708
GSPC	DeReFusion (reference)	0.06230	—
GSPC	revin-DLinear (local)	0.07007	0.8528
BTCUSD	DeReFusion (reference)	0.21797	—
BTCUSD	woLSTM (local)	0.22178	0.8734
BTCUSD	revin-DLinear (ref.)	0.22322	—

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VI. CONCLUSION AND NEXT STEPS

We executed a seven-run the component ablation of DeReFusion on GSPC and BTCUSD at $T=24$, seed 2021. Three conclusions follow.

- 1) The LSTM is not universally load-bearing. Removing it improves GSPC and degrades BTCUSD, so the residual branch’s value is asset-dependent.
- 2) The woDy variant is confounded by weight initialization and must not be read as a clean measure of the residual branch’s contribution. The fix is to restore the DLinear $1/L$ initialization in the ablation.
- 3) The Transformer-only residual is the best ablation on both assets, suggesting it is the more robust residual design at this data scale.

a) Next steps.: Re-run woDy with the DLinear initialization to obtain an unconfounded base-only number; extend the grid to the remaining horizons ($T \in \{1, 7, 12, 36\}$) and to the additional assets; and add a second seed to separate the two very close GSPC comparisons (woLSTM vs. full model, a 1.3% gap) from seed noise.

ACKNOWLEDGMENT

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REFERENCES

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